

# Independent Replication of “Quantum Algorithm Implementations for Beginners”

arXiv:1804.03719v3 — Abhijith J. et al., LANL, 2018/2022

Independent replication (QC-200 wave, 2026-07-05)

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## Verdict (one line)

**SPOT-CHECK.** All three independently chosen algorithms (Bernstein–Vazirani, Grover, Quantum Phase Estimation) reproduce their expected textbook outputs to statevector precision using a real Qiskit 2.5.0 statevector simulation, matching the paper’s qualitative description in Sections 2, 3, and 4.3. The paper is a 20-algorithm pedagogical survey with no single headline number, so this is a legitimate SPOT-CHECK (not full REPLICATED, which would require reproducing every algorithm plus every hardware-vs.-sim histogram in the paper).

## 1 Paper summary

Abhijith J. et al. (LANL, LA-UR-20-22353) is a 100+ page pedagogical review that presents 20 quantum algorithms with reference IBM Qiskit implementations, and for each algorithm compares outputs on the noiseless Qiskit simulator vs the noisy IBM Q 4/Q 5 hardware available in the 2018–2020 window. Algorithms covered include Grover’s search, Bernstein–Vazirani (BV), Simon’s algorithm, Deutsch–Jozsa, Quantum Phase Estimation (QPE), Shor’s factoring, HHL for linear systems, QAOA for MaxCut, VQE for small Hamiltonians, and several more.

The paper has no single “headline number”; it is a survey. Its verifiable content is therefore per-algorithm: each of the 20 sections gives an expected output shape (which measurement outcome should dominate) and a hardware histogram. A faithful independent reproduction picks a small subset of algorithms and verifies the paper’s expected *simulator-level* outcomes end-to-end.

## 2 Claims table

| Claim id | Statement                                                                                                                                                                       | Testable? |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| C1 (BV)  | For a linear oracle $f(x) = s \cdot x \pmod{2}$ with hidden $s \in \{0, 1\}^n$ , one query of the BV circuit recovers $s$ with probability 1 on a noiseless simulator (Sec. 3). | Yes       |

|                      |                                                                                                                                                                                                                                                                               |                                                              |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| C2 (Grover)          | For $N = 2^n$ and $M = 1$ marked item, after $k^* = \lfloor (\pi/4)\sqrt{N/M} \rfloor$ Grover iterations, the marked state has probability $\sin^2((2k^* + 1)\theta)$ with $\theta = \arcsin \sqrt{M/N}$ . For $n = 3$ , $M = 1$ : $k^* = 2$ and $P \approx 0.9453$ (Sec. 2). | Yes                                                          |
| C3 (QPE)             | For a 1-qubit unitary with eigenphase $\varphi = k/2^t$ exactly representable in $t$ counting qubits, QPE returns the integer $k$ with probability 1 (Sec. 4.3).                                                                                                              | Yes                                                          |
| C4 (Survey scope)    | The paper presents Qiskit-compatible reference implementations for 20 algorithms.                                                                                                                                                                                             | No (assertion of existence, not a numerical result)          |
| C5 (Hardware vs sim) | On the IBM Q4/Q5 hardware of 2018–2020, noiseless-sim outcomes differ measurably from hardware histograms.                                                                                                                                                                    | Yes, but requires IBM hardware access that is now restricted |

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## 3 Method

### 3.1 Environment

- Host: Darwin 25.3.0 (CherryRd), CPU only.
- Python 3.12.13 in `venv/`.
- Qiskit 2.5.0, qiskit-aer (via `pip install qiskit qiskit-aer numpy`), NumPy 2.5.1.
- Statevector via `qiskit.quantum_info.Statevector.from_instruction` — no shot noise, no sampler.

### 3.2 Bernstein–Vazirani (C1)

Hidden string  $s = 1011$  (MSB-first). Circuit uses  $n=4$  input qubits and 1 ancilla in  $|-\rangle$ . Oracle implemented by  $CX(x_i \rightarrow a)$  for each  $s_i = 1$ . Full script at `report/evidence/bv.py`. Runs in  $\ll 1$  s.

### 3.3 Grover (C2)

$n=3$ , marked state  $|101\rangle$  (decimal 5). Oracle: X-flip qubits where  $m$  has a 0, then  $H \cdot MCX \cdot H$  on the top qubit (implements a multi-controlled  $Z$ ), then undo the flips. Diffuser: standard  $HX(HMCXH)XH$  construction.  $k^* = 2$  iterations. Full script at `report/evidence/grover.py`.

### 3.4 QPE (C3)

Target  $U = \text{diag}(1, e^{2\pi i \varphi})$  with  $\varphi = 1/8$ , so  $2^t \varphi = 16 \cdot 1/8 = 2$  is an integer for  $t = 4$ . Eigenstate register prepared as  $|1\rangle$ . Counting register:  $t = 4$  qubits, controlled- $U^{2^j}$  from qubit  $j$ , then inverse QFT (explicit implementation with swaps + graded controlled phases). Full script at `report/evidence/qpe.py`.

## 4 Results vs paper

| Claim                | Paper prediction                                    | Our simulation                                                                                                 | Match?     |
|----------------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------|------------|
| C1 (BV, $s=1011$ )   | Recover $s$ with $P = 1$ in 1 query                 | Recover 1011 with $P = 0.999999999999986$ (i.e. 1 up to double precision)                                      | <b>YES</b> |
| C2 (Grover $N=8$ )   | After $k^* = 2$ : $P(\text{marked}) \approx 0.9453$ | $P( 101\rangle) = 0.9453124999999959$ (analytic $= 0.9453124999999999$ )                                       | <b>YES</b> |
| C3 ( $\varphi=1/8$ ) | (QPE $ 0010\rangle$ with $P = 1$ )                  | $ 0010\rangle$ with $P = 0.999999999999987$ ; all other basis states have probability $\leq 3 \times 10^{-31}$ | <b>YES</b> |

Full JSON outputs (with the entire probability distribution over all  $2^n$  basis states in each case) are saved to:

- `report/evidence/bv_result.json`
- `report/evidence/grover_result.json`
- `report/evidence/qpe_result.json`

## 5 Verdict + justification

**SPOT-CHECK.** All three independently chosen algorithms match their textbook / paper-stated simulator predictions to statevector precision (double). The paper is a survey with no single headline number, so full “REPLICATED” would require reproducing all 20 algorithms plus each hardware histogram; that was out of scope for this timed replication. What is claimed here is: the paper’s Qiskit-style pseudocode for these three canonical algorithms is faithful to the underlying quantum theory, and independent implementations (written from the paper’s descriptions without copying code) reproduce the promised noiseless outputs exactly.

## 6 Open Questions

See also `report/open_questions.json` for the machine-readable form.

- Q1 The paper reports per-algorithm hardware histograms from IBM Q4/Q5 chips (2018–2020). Those chips have since been retired. How would today’s `ibmq_qasm` equivalents (Falcon/Eagle) change the hardware-vs-sim gaps the paper documented, and can any of the “worked on hardware” verdicts be reversed by present-day noise floors?
- Q2 The QPE section uses  $t$  counting qubits with the implicit assumption that  $2^t\varphi$  is an integer for the paper’s demo. What is the probability of the second- and third-most-likely outcomes when  $\varphi$  is *not* representable exactly in  $t$  bits? A parameter sweep would quantify how the paper’s “exact” demo degrades to the generic  $1/(k^2\pi^2)$ -tailed distribution, which the paper does not plot.

- Q3 The Grover section quotes success probability  $> 1/2$  as “the guarantee” but the actual  $k^* = \lfloor (\pi/4)\sqrt{N/M} \rfloor$  rounding can either over- or under-shoot the optimal angle. Which specific  $(N, M)$  pairs land closer to  $1/2$  than to  $1$ ? This has practical consequence for the paper’s IBM-hardware runs but is not tabulated in the paper.
- Q4 The paper’s Bernstein–Vazirani implementation uses  $n+1$  qubits and 1 oracle call, achieving  $P = 1$  on a simulator. The paper does not quantify the crossover point at which BV becomes hardware-*worse* than classical bit-by-bit probing under a given depolarizing-noise rate. A depolarizing sweep across  $n = 2 \dots 16$  would answer whether the paper’s “quantum advantage” argument survives on today’s noisiest public chips.
- Q5 The paper’s own extraction (Marker on the same PDF) regularly bleeds the first author’s surname “ABHIJITH” into the title. This is a corpus-hygiene issue that likely affects downstream tools using the paper’s Marker parse as ground truth. What fraction of the paper’s cited references would be mis-attributed if a metadata pipeline used the Marker-parsed title as the paper’s true title?